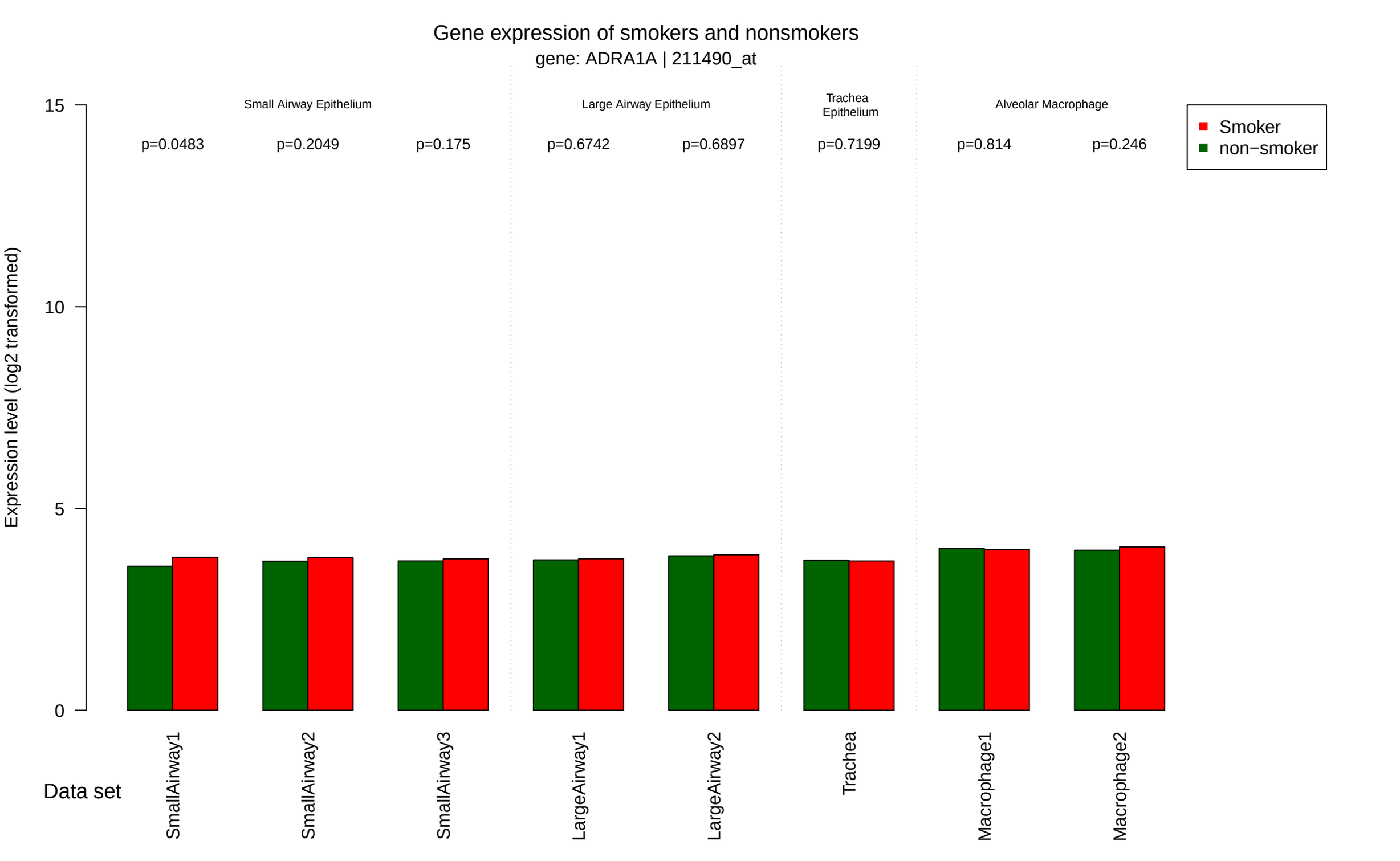
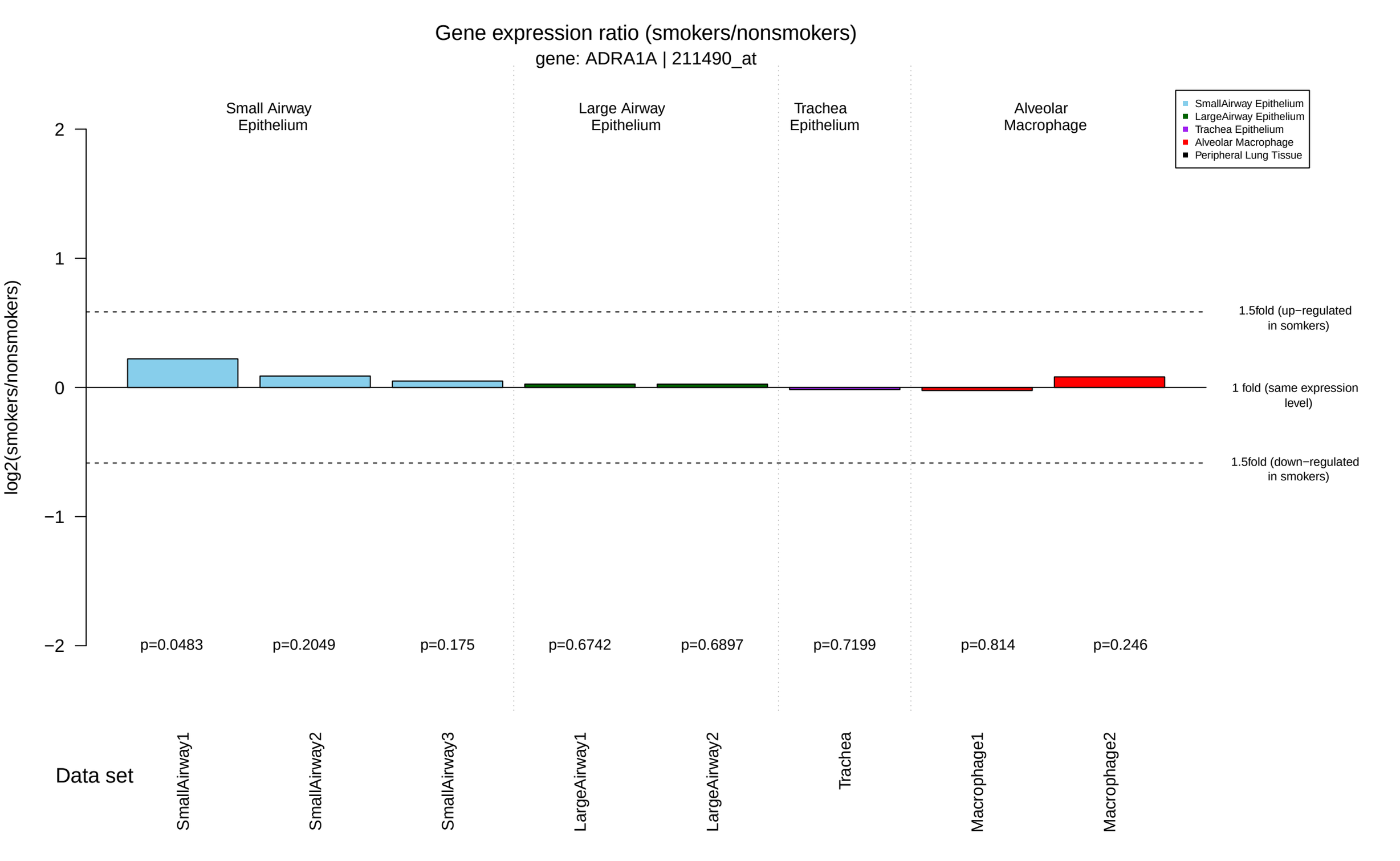
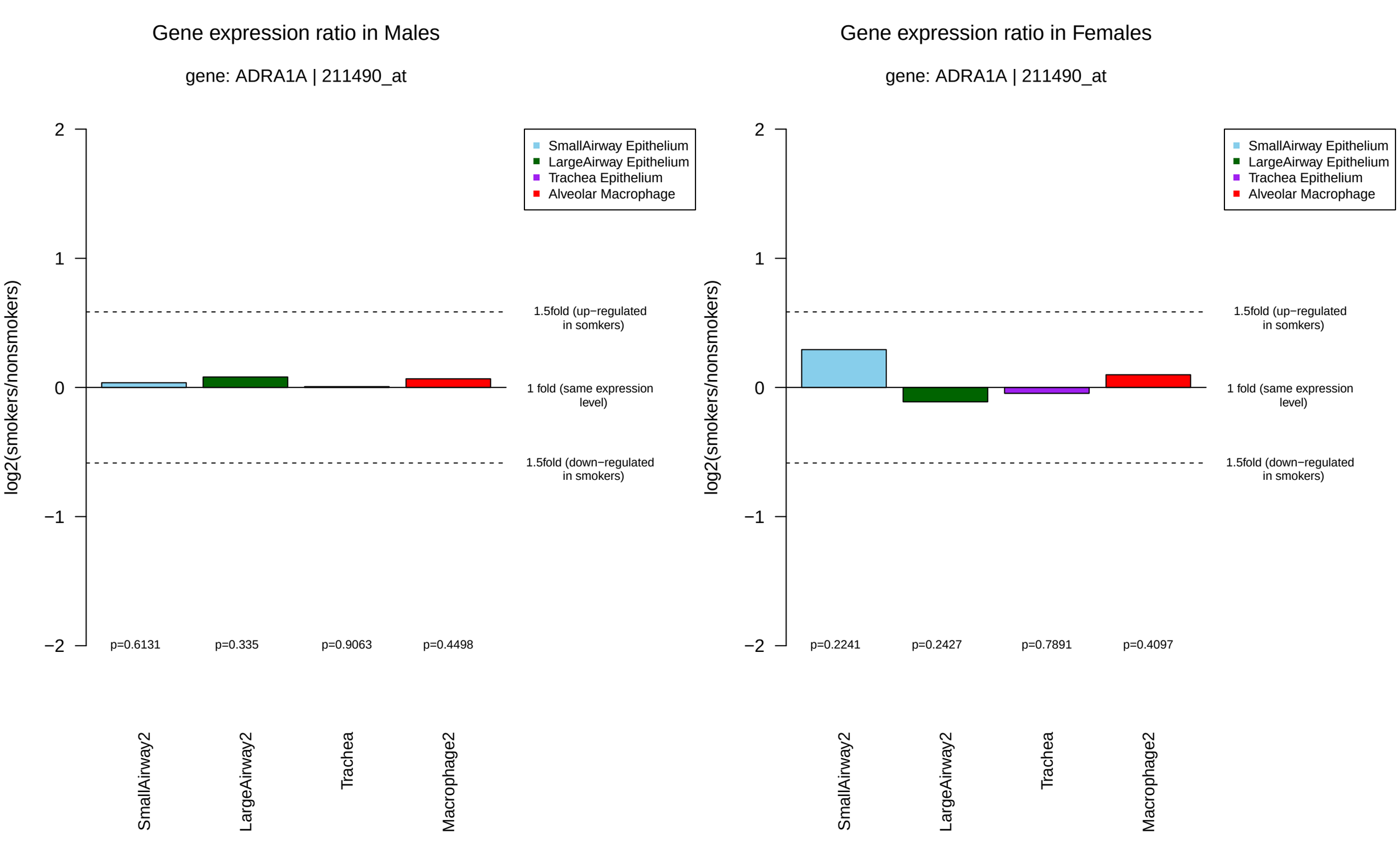
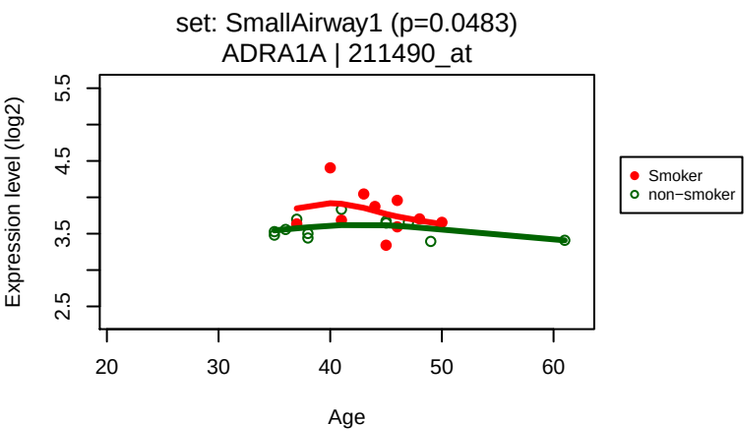
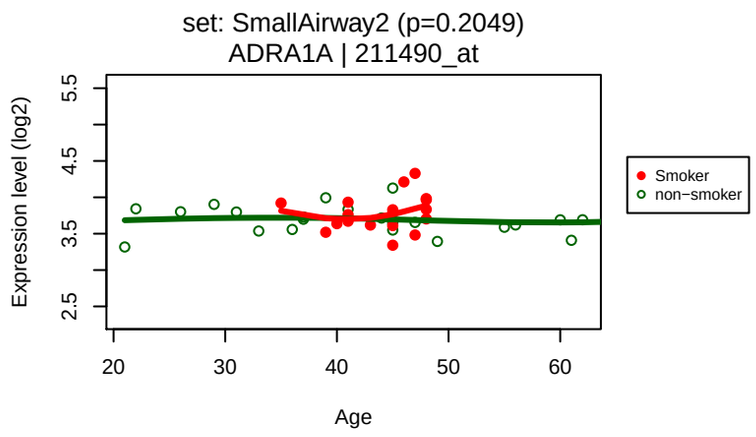
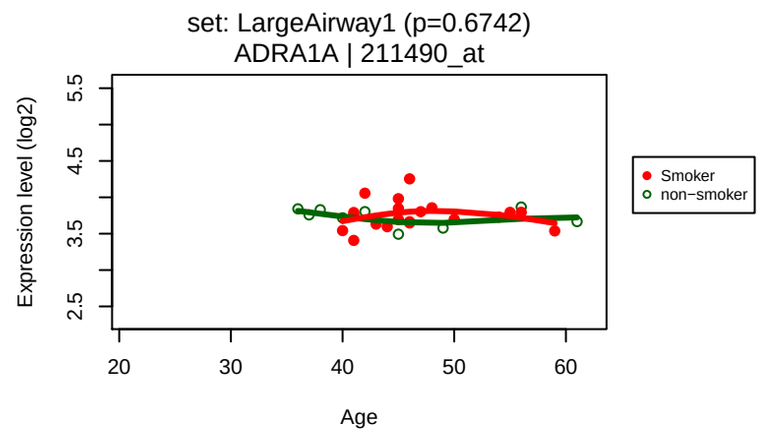

Gene expression of smokers and nonsmokers
gene: ADRA1A | 211490_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0483
p=0.2049
p=0.175
p=0.6742
p=0.6897
p=0.7199
p=0.814
p=0.246
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

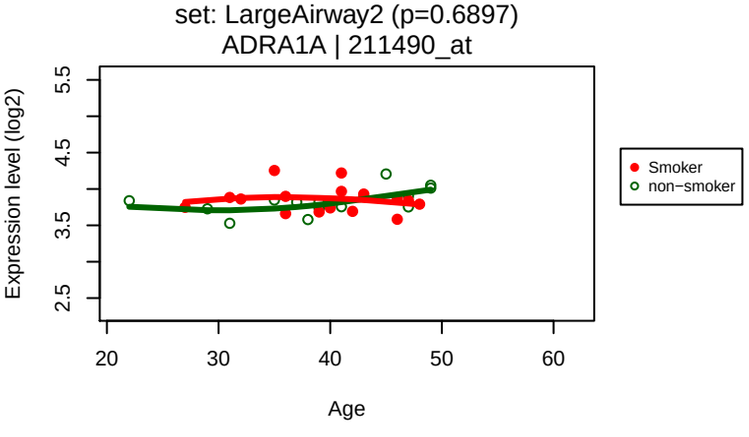

Gene expression ratio (smokers/nonsmokers)
gene: ADRA1A | 211490_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0483
p=0.2049
p=0.175
p=0.6742
p=0.6897
p=0.7199
p=0.814
p=0.246
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

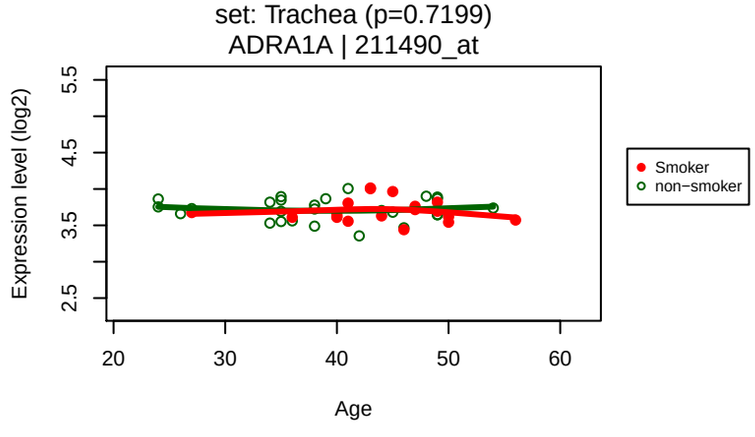

Gene expression ratio in Males
gene: ADRA1A | 211490_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6131
p=0.335
p=0.9063
p=0.4498
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADRA1A | 211490_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2241
p=0.2427
p=0.7891
p=0.4097
SmallAirway2
LargeAirway2
Trachea
Macrophage2

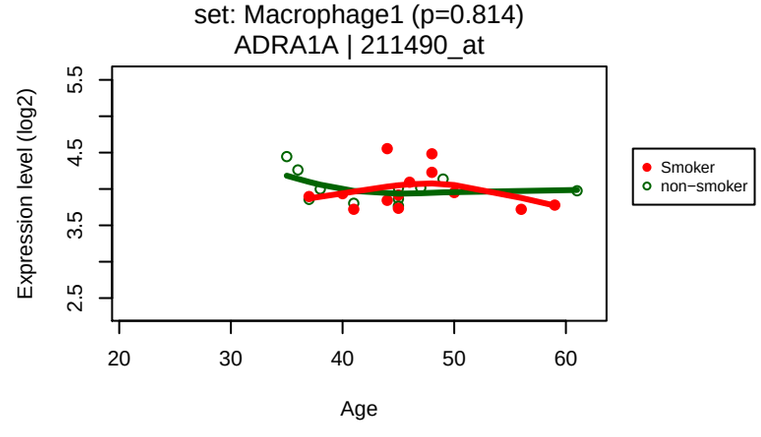

set: SmallAirway1 (p=0.0483)
ADRA1A | 211490_at
Expression level (log2)
Age
Smoker
non−smoker

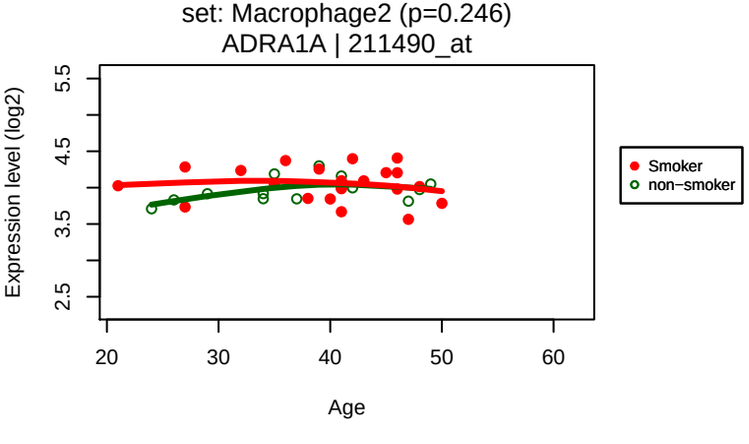

set: SmallAirway2 (p=0.2049)
ADRA1A | 211490_at
Expression level (log2)
Age
Smoker
non−smoker

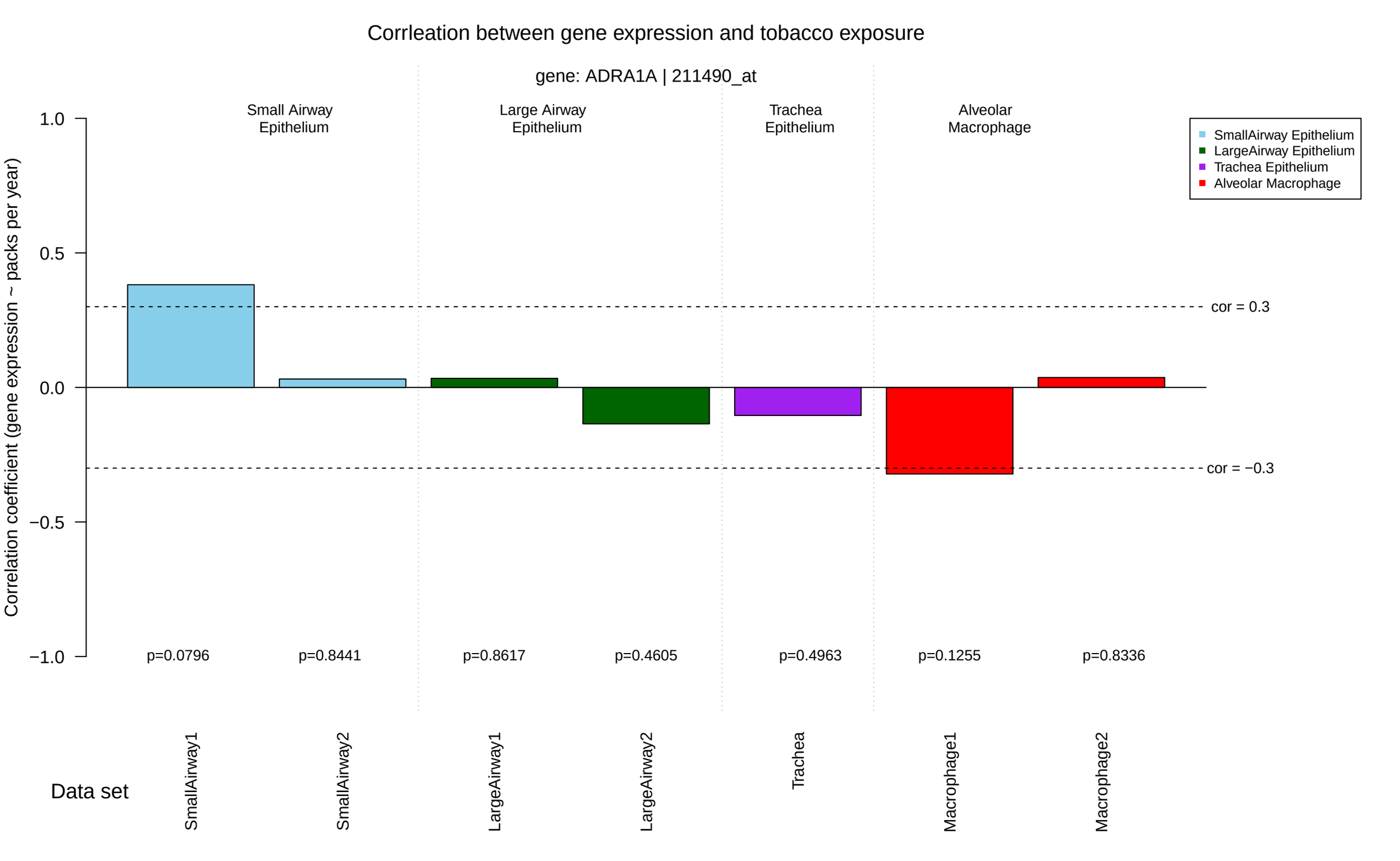

set: LargeAirway1 (p=0.6742)
ADRA1A | 211490_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6897)
ADRA1A | 211490_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7199)
ADRA1A | 211490_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.814)
ADRA1A | 211490_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.246)
ADRA1A | 211490_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADRA1A | 211490_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0796
p=0.8441
p=0.8617
p=0.4605
p=0.4963
p=0.1255
p=0.8336
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2